

Rhea Data Science Internship Assessment

Background

Rhea provides agricultural recommendations to farmers based on factors such as:

- Soil properties
- Climate conditions
- Location
- Crop performance

To improve recommendation quality, we need high-quality agricultural datasets.

Your task is to source, clean, analyze, and prepare a dataset that could be useful for building agricultural recommendation systems.

Task 1: Data Sourcing (20 Marks)

Find and download a publicly available agricultural dataset.

Examples include:

- Crop yield datasets
- Soil nutrient datasets
- Weather datasets
- Fertilizer recommendation datasets
- Crop recommendation datasets

Possible sources:

- Kaggle
- FAO
- World Bank Open Data
- Government agricultural portals
- Research datasets

Deliverables

Provide:

- Dataset source

- Dataset description
- Why you selected it
- How it could improve agricultural recommendations

Task 2: Data Cleaning (25 Marks)

Perform data cleaning on the selected dataset.

Examples:

- Handle missing values
- Remove duplicates
- Fix inconsistent entries
- Correct data types
- Identify outliers

Deliverables

Provide:

- Summary of issues found
- Cleaning steps taken
- Cleaned dataset

Task 3: Exploratory Data Analysis (25 Marks)

Perform EDA and generate insights.

Examples:

- Distribution plots
- Correlation analysis
- Feature relationships
- Missing-value analysis

Deliverables

At least:

- 5 visualizations

- 5 insights

Example Questions

- Which soil nutrients vary the most?
- Which crops appear most frequently?
- Are there relationships between rainfall and crop suitability?
- Which features might influence recommendations?

Task 4: Recommendation Feature Engineering (20 Marks)

Imagine the dataset will be used in Rhea's recommendation system.

Create at least 3 new features that could improve recommendations.

Examples:

Original Features	New Feature
Nitrogen, Phosphorus, Potassium	NPK Score
Rainfall, Temperature	Growing Condition Index
pH	Soil Suitability Category

Deliverables

Explain:

- Feature formula
- Why it is useful
- Expected impact on recommendations
- Practicality

Task 5: Recommendation Logic (10 Marks)

Design a simple recommendation approach.

Example:

Given:

- Nitrogen = 85 ppm
- Phosphorus = 40 ppm
- Potassium = 50 ppm
- pH = 6.5
- Rainfall = 120 mm

How would you recommend a crop?

You may use:

- Rule-based logic
- Similarity matching
- Simple machine learning model

Deliverables

A one-page explanation.

NOTE: No advanced modeling is required.

Bonus Task (+10 Marks)

Use an LLM (ChatGPT, Claude, Gemini, etc.) to generate additional insights from the dataset.

Explain:

- Prompt used
- Insights obtained
- Whether the insights were useful

This tests modern AI-assisted analytics skills.

Submission Requirements

Submit to Info@rhea.africa

1. Jupyter Notebook (.ipynb)
2. Cleaned Dataset
3. PDF Report (maximum 5 pages)

4. README.md

Scoring Rubric

Area	Marks
Data Sourcing	20
Data Cleaning	25
EDA	25
Feature Engineering	20
Recommendation Logic	10
Bonus	10
Total	100 (+10 Bonus)